

Linear Algebra

Solutions to Practice Problems

Chapter 2, Part 2 — Rank, Existence & Uniqueness, Invertible Matrices, Determinants

How This Pairs With Your Notes

This is the companion answer key to *LinearAlgebra_Ch2_Part2_Notes.pdf*. Each problem is re-stated in an orange box (so this file is self-contained), followed by a full worked solution in teal. Try each problem yourself first – the value of a practice problem comes from attempting it before you check the answer.

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1 Solutions – §2.5 (Rank of a Matrix)

Problem 2.5.1

Find rank $\begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{pmatrix}$.

Solution

Every row is a multiple of $(1, 2)$: row 2 = $2 \times$ row 1, row 3 = $3 \times$ row 1. Row-reducing leaves only one nonzero row ($R_{21}(-2)$, $R_{31}(-3)$ both wipe out entirely). So rank = **1**.

Problem 2.5.2

Find rank $\begin{pmatrix} 1 & 1 & 2 \\ 2 & 2 & 4 \\ 1 & 0 & 1 \end{pmatrix}$.

Solution

Row 2 = $2 \times$ row 1, so $R_{21}(-2)$ wipes row 2 to zero: $\begin{pmatrix} 1 & 1 & 2 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$. Row 3 is *not* a multiple of row 1 (compare $(1, 0, 1)$ against $(1, 1, 2)$ – the middle entries alone rule out proportionality). Swapping

row 3 up and continuing, two nonzero rows survive. rank = **2**.

Problem 2.5.3

Give an example of a 3×3 matrix with rank exactly 1, and one with rank exactly 0.

Solution

Rank 1: $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix}$ (every row is a multiple of $(1, 2, 3)$, so only one independent direction survives). Rank 0: the zero matrix $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ (no nonzero rows at all – the *only* matrix with rank 0).

Problem 2.5.4

Can a 2×5 matrix have rank 3? Explain why or why not.

Solution

No. A 2×5 matrix has only 2 rows to begin with, so after row reduction there can never be more than 2 nonzero rows – rank 3 would require 3 independent rows out of only 2 available. In general, $\text{rank}(A) \leq (\text{number of rows of } A)$, always.

Problem 2.5.5 (Challenge)

If $\text{rank}(A) = r$ for an $m \times n$ matrix A , what is the largest possible value of r ?

Solution

$r \leq \min(m, n)$. The row-reduction argument in Problem 2.5.4 shows $r \leq m$ (can't have more nonzero rows than actual rows); and since row rank equals column rank (Proposition 3.3.20 – coming up in Chapter 3), the same argument on columns gives $r \leq n$ too. Combining both bounds, the largest possible rank is $r = \min(m, n)$, achieved when the matrix has “no wasted rows or columns” at all (this is called **full rank**).

2 Solutions – §2.6 (Existence & Uniqueness of Solutions)

Problem 2.6.1

For $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ and $b = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$, compute $\text{rank}(A)$ and $\text{rank}([A \ b])$, then state how many solutions $Ax = b$ has.

Solution

$\text{rank}(A) = 1$ (row 2 = 2×row 1). For $[A \ b] = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$: row 2 = 2×row 1 here too (the augmented column keeps the same proportionality), so $\text{rank}([A \ b]) = 1$ as well. Here $r = r_a = 1 < n = 2$: by Theorem 2.6.1 case (1), **infinitely many solutions**.

Problem 2.6.2

Repeat Problem 2.6.1 with $b = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$ instead.

Solution

$\text{rank}(A) = 1$, unchanged. But now $[A \ b] = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \end{pmatrix}$: $R_{21}(-2)$ gives $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{pmatrix}$, a nonzero row surviving, so $\text{rank}([A \ b]) = 2$. Here $r = 1 < r_a = 2$: by Theorem 2.6.1 case (3), **no solution**. (Same coefficient matrix as Problem 2.6.1 – only the right-hand side changed, and that alone flips “infinite” to “none.”)

Problem 2.6.3

For a 4×4 system with $\text{rank}(A) = 4$, how many solutions does $Ax = b$ have, for any choice of b ?

Solution

Here $n = 4$ and $r = \text{rank}(A) = 4 = n$. Since $[A \ b]$ has only 4 rows, $r_a = \text{rank}([A \ b]) \leq 4$; but $r_a \geq r = 4$ always. So $r_a = 4 = r = n$ automatically, for *every* choice of b – Theorem 2.6.1 case (2) applies every time: **a unique solution, for any b whatsoever**. (This is A being invertible, previewing §2.7.)

Problem 2.6.4

Does $x - y + 2z = 0$, $2x - 2y + 4z = 0$ have a non-trivial solution?

Solution

Coefficient matrix $A = \begin{pmatrix} 1 & -1 & 2 \\ 2 & -2 & 4 \end{pmatrix}$; row 2 = $2 \times$ row 1, so $\text{rank}(A) = 1$. Here $n = 3$ (three unknowns) and $\text{rank}(A) = 1 < 3 = n$. By Corollary 2.6.3, **yes** – a non-trivial solution exists (in fact infinitely many: 2 free variables’ worth).

Problem 2.6.5 (Challenge)

Explain why a non-square matrix A with more unknowns than equations can never have $Ax = b$ uniquely solvable for every b .

Solution

Say A is $m \times n$ with $m < n$ (fewer equations than unknowns). Then $\text{rank}(A) \leq m < n$ always (rank can’t exceed the number of rows) – so $\text{rank}(A) < n$ unconditionally. By Corollary 2.6.3, this means $Ax = 0$ *always* has a non-trivial solution. But Proposition 2.6.4 says “ $Ax = b$ uniquely solvable for every b ” and “ $Ax = 0$ has a non-trivial solution” can never both hold. Since we just showed the second statement is always true here, the first can never be: **more unknowns than equations rules out unique solvability for every b** , before you even look at the specific numbers in A .

3 Solutions – §2.7 (Invertible Matrices)

Problem 2.7.1

Use Gauss–Jordan elimination to find the inverse of $\begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$.

Solution

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 1 & 3 & 0 & 1 \end{array} \right] \xrightarrow{R_{21}(-1)} \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & -1 & 1 \end{array} \right] \xrightarrow{R_{12}(-2)} \left[\begin{array}{cc|cc} 1 & 0 & 3 & -2 \\ 0 & 1 & -1 & 1 \end{array} \right].$$

So $A^{-1} = \begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix}$.

Problem 2.7.2

Verify your answer to Problem 2.7.1 by checking $AA^{-1} = I_2$ directly.

Solution

$$\begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1(3) + 2(-1) & 1(-2) + 2(1) \\ 1(3) + 3(-1) & 1(-2) + 3(1) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2. \checkmark$$

Problem 2.7.3

Is $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ invertible? Justify.

Solution

$\det \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = 1(4) - 2(2) = 0$. By Corollary 2.8.16 (equivalently, since row 2 is a multiple of row 1, $\text{rank} = 1 \neq 2$, failing Theorem 2.7.7’s full-rank condition), **this matrix is not invertible**.

Problem 2.7.4

If A, B are both invertible $n \times n$ matrices, simplify $(AB)^{-1}B$.

Solution

By Theorem 2.7.4, $(AB)^{-1} = B^{-1}A^{-1}$. So

$$(AB)^{-1}B = (B^{-1}A^{-1})B = B^{-1}A^{-1}B.$$

This is as far as it simplifies in general – it collapses further (to B^{-1}) only in the special case that $A^{-1}B = B$, i.e. $A = I_n$. The point of the exercise is applying the “reverse the order” rule correctly, not expecting extra cancellation that isn’t actually there.

Problem 2.7.5

Find the inverse of $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ using Gauss–Jordan elimination, showing every step.

Solution

$$\begin{aligned}
& \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_{31}(-1)} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & -1 & 0 & 1 \end{array} \right] \\
& \xrightarrow{R_{32}(-1)} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -2 & -1 & -1 & 1 \end{array} \right] \xrightarrow{R_3(-1/2)} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array} \right] \\
& \xrightarrow{\substack{R_{13}(-1) \\ R_{23}(-1)}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array} \right].
\end{aligned}$$

So $A^{-1} = \begin{pmatrix} 1/2 & -1/2 & 1/2 \\ -1/2 & 1/2 & 1/2 \\ 1/2 & 1/2 & -1/2 \end{pmatrix}$. (Quick sanity check: the matrix is symmetric, and so is its inverse – exactly as it should be, since $(A^{-1})^t = (A^t)^{-1} = A^{-1}$ whenever A itself is symmetric.)

Problem 2.7.6 (Challenge)

If A is invertible and $A^2 = A$, show that $A = I_n$.

Solution

Since A is invertible, A^{-1} exists. Multiply both sides of $A^2 = A$ on the left by A^{-1} :

$$A^{-1}A^2 = A^{-1}A \implies (A^{-1}A)A = I_n \implies I_n A = I_n \implies A = I_n.$$

(This also explains why the idempotent examples from Chapter 1, like $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, are *never* invertible except in the trivial case $A = I$ – idempotence and invertibility only coexist at the identity.)

4 Solutions – §2.8 (Determinants & Cramer's Rule)**Problem 2.8.1**

Compute $\det \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$.

Solution

$$\det = 3(4) - 1(2) = 12 - 2 = \mathbf{10}.$$

Problem 2.8.2

Compute $\det \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix}$ by cofactor expansion along the first row.

Solution

$$\begin{aligned}
\det(A) &= 1 \det \begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix} - 2 \det \begin{pmatrix} 0 & 3 \\ 2 & 1 \end{pmatrix} + 0 \det \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} \\
&= 1(1 - 6) - 2(0 - 6) + 0 = 1 - 2(-6) = 1 + 12 = \mathbf{13}.
\end{aligned}$$

Problem 2.8.3

Recompute the determinant in Problem 2.8.2 by expanding along a different row or column, and check you get the same answer.

Solution

Expanding along column 1 (entries 1, 0, 2) instead:

$$\begin{aligned}\det(A) &= 1 \det \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} - 0 \det \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} + 2 \det \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix} \\ &= 1(1) - 0 + 2(6 - 0) = 1 + 12 = \mathbf{13}.\end{aligned}$$

Matches Problem 2.8.2 exactly, as Remark 2.8.8 guarantees.

Problem 2.8.4

Is $\begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$ singular or non-singular?

Solution

$\det = 2(2) - 4(1) = 4 - 4 = 0$. Since $\det = 0$, the matrix is **singular** (and hence, by Corollary 2.8.16, also not invertible).

Problem 2.8.5

Find $\text{Adj}(A)$ and A^{-1} for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ using the determinant/adjoint formula.

Solution

$\det(A) = 1(4) - 2(3) = -2$. For a 2×2 matrix, the cofactors are single entries with alternating signs: $C_{11} = 4$, $C_{12} = -3$, $C_{21} = -2$, $C_{22} = 1$. The adjoint is the *transpose* of the cofactor matrix:

$$\text{Adj}(A) = \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix}, \quad A^{-1} = \frac{1}{\det(A)} \text{Adj}(A) = \frac{1}{-2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix}.$$

Direct check via the familiar 2×2 shortcut $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ gives the identical answer. ✓

Problem 2.8.6 (Challenge)

Use Cramer's rule to solve $x + y = 5$, $2x - y = 1$.

Solution

$A = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}$, $b = \begin{pmatrix} 5 \\ 1 \end{pmatrix}$, $\det(A) = 1(-1) - 1(2) = -3$. Replacing each column by b in turn:

$$\det(A_1) = \det \begin{pmatrix} 5 & 1 \\ 1 & -1 \end{pmatrix} = -5 - 1 = -6, \quad \det(A_2) = \det \begin{pmatrix} 1 & 5 \\ 2 & 1 \end{pmatrix} = 1 - 10 = -9.$$

So $x = \frac{\det(A_1)}{\det(A)} = \frac{-6}{-3} = 2$, and $y = \frac{\det(A_2)}{\det(A)} = \frac{-9}{-3} = 3$. **Solution:** $(x, y) = (2, 3)$. Check by substitution: $x + y = 2 + 3 = 5$ ✓; $2x - y = 4 - 3 = 1$ ✓.

That covers every practice problem for Chapter 2, Part 2. Ask for Chapter 3's notes and solutions whenever you're ready to continue.